

[ACJC/2024/Q4]

4 Name your Jupyter Notebook as:

TASK4_<your name>_<centre number>_<index number>.ipynb

A supermarket keeps records about its customers, items and purchases. The supermarket wants to create a suitable database to store the data and allow them to run searches for specific data. The database will have three tables: a table to store data about the items, a table about the customers, and a table about the purchases. The fields in each table are:

Item:

- ItemID – unique item number, for example, 1023
- Name – the name of the item
- Price – the price of the item in cents, for example, 350 for \$3.50

Customer:

- CustomerNumber – customer's unique number, for example, 456
- Name – customer's name

Purchase:

- PurchaseID – the purchase's unique ID, for example, 12
- CustomerNumber – the customer's unique number
- ItemID – the unique item number
- Quantity – the number of that item purchased
- DateTime – the date and time of the purchase in YYYYMMDD format.

For each of the sub-tasks 4.1 to 4.3, add a comment statement at the beginning of the code, using the hash symbol '#' to indicate the sub-task the program code belongs to, for example:

```
In [1]: #Task 4.1
        Program code
        Output:
```

Task 4.1

Write a Python program that uses SQL code to create the database `SUPERMARKET` with the three tables given. Define the primary and foreign keys for each table. [6]

Task 4.2

The text files `ITEM.txt`, `CUSTOMER.txt` and `PURCHASE.txt` contains the comma-separated values for each of the tables in the database.

Write a Python program to read the data from each file and then store each item of data in the correct place in the database. [5]

Task 4.3

Write a Python program to input a customer's number and date and return the names and quantities of all the items that they have purchased on or after that date. [6]

Test your program by running the application with the member number 200 and date 20240102. [2]

Save your Jupyter Notebook.

[ACJC/2024/Q4 Cont']

Task 4.4

Write a Python program and the necessary files to create a web application that displays the following data for all purchases made in the year 2024.

- Customer name
- Item name
- Quantity purchased

The program should return an HTML document that enables the web browser to display a table with the required data.

Save your program code as

TASK4_4_<your name>_<centre number>_ <index number>.py

with any additional files/subfolders as needed in a folder named

TASK4_4_<your name>_<centre number>_ <index number>

[6]

Run the web application.

Save the webpage output as:

TASK4_4_<your name>_<centre number>_ <index number>.html

[2]